

 SLIIT <i>Discover Your Future</i>	DEPARTMENT OF INFORMATION TECHNOLOGY		
	FACULTY OF COMPUTING		

MODULE OUTLINE					
Module Name	Theory and Practices in Statistical Modeling				
Module Code	IT3011	Version No.	2017	-	1
Year	3	Semester	1		
Credit Points	4				
Pre-requisites	None				
Co-requisites	None				
Methods of Delivery		Lectures (Face-to-face)	2	Hours/Week	
		Tutorials	1	Hours/Week	
		Labs	2	Hours/Week	
Course Web Site		http://courseweb.sliit.lk/			
Date of Original Approval		November, 2017			
Date of Next Review		November, 2022			

MODULE DESCRIPTION	
Introduction	The primary objective of this course is to introduce commonly used underlying statistical theory and practices of real-life analytical and predictive solutions. Topics are covering from fundamentals of statistics to specific details of some modelling theories. R studio statistical software is used in teaching and assessments.
Learning Outcomes	At the end of the module student will be able to:
	LO1: Distinguish characteristics of discrete and continuous probability distributions and use them in statistical modeling.
	LO2: Apply the theory of hypothesis testing for real world datasets leading to managerial decisions.
	LO3: Apply and interpret outcome of the Regression methods in the context of experimental environment.

	LO4:	Analyze datasets using the exponential family of distributions that includes the Normal, Binomial, Poisson and the Gamma distributions.		
	LO5:	Use the Time series techniques for scientific forecasting.		
	LO6:	Demonstrate Bayesian methods can be used to solve real world problems and communicate outcome to a diverse audience		
Assessment Criteria	During the semester, there will be a mid-term test, assignments, and a final examination. The mid-term test will be based on the practical work, the questions discussed in tutorial sessions, and lecture material covered until the week before it is held. Few assignments will be given to assess the certain topics during the semester. The final examination will be a comprehensive written examination based on the topics covered during the semester.			
	Continuous Assessments			
	• Midterm Examination	25	%	LO1- LO2
	• Assignments	25	%	LO1- LO6
	End Semester Assessment			
	• Final Examination	50	%	LO1 – LO6
	TOTAL	100	%	
Estimated Student Workload	Contact Hours			
	• Lecture	26 hours		
	• Tutorial	13 hours		
	• Laboratory	26 hours		
	Time Allocated for Assessments			
	• Continuous Assessments	1 hours		
	• Final Examination	2 hours		
	Reading and Independent Study	132 hours		
	TOTAL	200 hours		
Module Requirement	To pass this module, students need to obtain a pass mark in both “Continuous Assessments” and “End of the Semester Examination” components which would result in an overall mark that would qualify for a “C” grade or above.			
Primary References	1. Hogg R.V., McKean J.W. and Craig A., Introduction to Mathematical Statistics, 8 th Edition, 2019.			
	2. Provost F. and Fawcett T., Data Science for Business, O’Reilly Media, Inc, 2013.			
	3. EMC Education Services, Data Science and Big Data Analytics – Discovering, Analyzing, Visualizing and Presenting Data, John Wiley & Sons, Inc, 2015.			

CONTENTS OF THE MODULE	
Topics	Learning Outcomes covered
1. Probability Distributions - Discrete Probability Distributions (Hypergeometric Distributions, Geometric Distributions, Negative Binomial Distributions) - Continuous Probability Distributions (Gamma Distributions, Chi-square Distributions, Beta Distributions)	LO1, LO4
2. Hypothesis Testing - Hypothesis Tests for Proportions (Testing One Population Proportion, Testing Two Population Proportions) - Hypothesis Tests for Variances (Testing One Population Variance, Testing Two Population Variances) - Comparison of two population means (Two Independent Populations, Paired Samples)	LO2
3. Multiple Linear Regression - Regression diagnostics and addressing violations of regression assumptions (Residual Analysis as a regression diagnostics, Lack of Fit and Pure Error, Solutions to Model Departures) - Multiple Linear Regression Model (Inference for individual regression coefficients, Test for addition of a single variable / group of variables, Partial R-squared, Adjusted R-squared, Variable Selection and Model Building)	LO3
4. Time Series Analysis - Stationarity - Volatility - Time series Models (ARIMA, VAR, GARCH)	LO5
5. Bayesian Analysis - Applying Bayes' Rule to Data Science - Conditional Independence and Naive Bayes - Advantages and Disadvantages of Naive Bayes	LO6
GENERIC INFORMATION	

Any type of plagiarism is not allowed.

Plagiarism: Academic honesty is crucial to a student's credibility and self-esteem, and ultimately reflects the values and morals of the Institute as whole. A student may work together with one or a group of students discussing assignment content, identifying relevant references, and debating issues relevant to the subject. Plagiarism occurs when the work of another person, or persons, is used and presented as one's own.

-----End of Module Outline-----